CS 210: Worksheet 9 *Dr. Mudassir Shabbir, September 24, 2026*

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Mathematical induction. Questions from Rosen, *Discrete Mathematics and Its Applications*, 7th ed., Section 5.1, pp. 329–332. Original exercise numbers are retained. For induction proofs, state the basis, hypothesis, and step, and identify where the hypothesis is used. For Exercise 49, identify the error.

Exercise 3. Let $P(n)$ be the following statement for a positive integer n.

$$1^2 + 2^2 + \cdots + n^2 = \frac{n(n+1)(2n+1)}{6}$$

(a) What is the statement $P(1)$?

(b) Show that $P(1)$ is true, completing the basis step.

(c) What is the inductive hypothesis?

(d) What do you need to prove in the inductive step?

(e) Complete the inductive step, identifying where you use the inductive hypothesis.

Exercise 6. Use mathematical induction to prove the identity for every positive integer n. Identify where you use the inductive hypothesis.

$$1 \cdot 1! + 2 \cdot 2! + \cdots + n \cdot n! = (n+1)! - 1$$

Exercise 9.

(a) Find a formula for the sum of the first n even positive integers.

(b) Use mathematical induction to prove your conjecture from part (a). Identify where you use the inductive hypothesis.

Exercise 25. Let h be a real number with $h > -1$. Use mathematical induction to prove the inequality for every nonnegative integer n.

$$1 + nh \leq (1 + h)^n$$

Exercise 31. Use mathematical induction to prove that 2 divides $n^2 + n$ whenever n is a positive integer.

$$2 \mid (n^2 + n)$$

Exercise 38. Let $A_1, \ldots, A_n$ and $B_1, \ldots, B_n$ be sets with $A_j \subseteq B_j$ for $j = 1, \ldots, n$. Use mathematical induction to prove:

$$\bigcup_{j=1}^{n} A_j \subseteq \bigcup_{j=1}^{n} B_j$$

Here $\bigcup_{j=1}^{n} A_j = A_1 \cup \cdots \cup A_n$; $A \subseteq B$ means every element of A is in B.

Exercise 49. What is wrong with this purported proof that all horses are the same color?
Claim and base. Let $P(n)$ say that all horses in any set of n horses have the same color. Clearly, $P(1)$ is true.
Purported inductive step. Assume $P(k)$. Number any $k + 1$ horses from 1 to $k + 1$. The first k horses share one color, and the last k share one color.
Purported conclusion. The proof claims that these two groups overlap, so all k+1 horses have the same color. Thus $P(k$+1) holds.

Exercise 54. Use mathematical induction to show that, given a set of $n+1$ positive integers, none exceeding $2n$, at least one integer in the set divides another integer in the set.
The two integers must be distinct; n is a positive integer.

Proof. Let $P(n)$ be the statement above. We prove $P(n)$ for all $n \geq 1$ by mathematical induction.

Base case ($n = 1$). We need a set of 2 positive integers, each at most 2. The only positive integers at most 2 are 1 and 2, and a set has no repeated elements, so $S = \{1, 2\}$. Since $1 \mid 2$ and $1 \neq 2$, $P(1)$ is true.

Inductive step. Assume $P(n)$ holds for some fixed $n \geq 1$; that is, every set of $n+1$ positive integers from $\{1, 2, \ldots, 2n\}$ contains two distinct elements, one of which divides the other (the *inductive hypothesis*).
Let S be any set of $(n+1)+1 = n+2$ positive integers, each at most $2(n+1) = 2n+2$. We must find distinct $a, b \in S$ with $a \mid b$. The only elements of S that can lie outside $\{1, \ldots, 2n\}$ are $2n+1$ and $2n+2$, so we consider how many of these S contains.

Case 1: S contains at most one of $2n+1$ and $2n+2$. Removing at most one element from S leaves at least $(n+2)-1 = n+1$ elements, all in $\{1, \ldots, 2n\}$. Choose any $n+1$ of them. By the inductive hypothesis, two distinct elements among these satisfy $a \mid b$. Both lie in S, so we are done.

Case 2: S contains both $2n+1$ and $2n+2$.

Subcase 2a: $n+1 \in S$. Since $2n+2 = 2(n+1)$, we have $(n+1) \mid (2n+2)$. These numbers are distinct because $n \geq 1$ gives $2n+2 > n+1$. Both lie in S, so we are done.

Subcase 2b: $n+1 \notin S$. Remove $2n+1$ and $2n+2$ from S; the remaining n elements all lie in $\{1, \ldots, 2n\}$. Now add $n+1$ to form

$$T = \big(S \setminus \{2n+1,\, 2n+2\}\big) \cup \{n+1\}.$$

Observe:

1. T has exactly $n+1$ elements, because $n+1 \notin S$, so adding it creates no duplicate.
2. Every element of T is at most $2n$: this holds for the n elements kept from S, and $n+1 \leq 2n$ because $n \geq 1$.
3. Every element of T other than $n+1$ belongs to S.

By the inductive hypothesis, there are distinct $a, b \in T$ with $a \mid b$.

- *Neither a nor b equals $n+1$.* Then $a, b \in S$, and we are done.
- $b = n+1$. Then $a \mid (n+1)$ and $(n+1) \mid (2n+2)$, so by transitivity of divisibility, $a \mid (2n+2)$. Here $a \in S$ since $a \neq n+1$, and $2n+2 \in S$ by the assumption of Case 2. They are distinct because $a \leq 2n < 2n+2$. Thus a and $2n+2$ are the required pair in S.
- $a = n+1$. Then $(n+1) \mid b$ with $b \neq n+1$. The positive multiples of $n+1$ are $n+1$, $2(n+1)$, $3(n+1), \ldots$, so $b \geq 2(n+1) = 2n+2$. But every element of T is at most $2n$, a contradiction. Hence this situation cannot occur.

In every possible situation, S contains distinct elements a, b with $a \mid b$, so $P(n+1)$ holds.

Conclusion. Since $P(1)$ is true and $P(n) \Rightarrow P(n+1)$ for every $n \geq 1$, the principle of mathematical induction gives that $P(n)$ holds for all positive integers n. □

Exercise 55. A knight moves one square horizontally and two vertically, or two horizontally and one vertically, with either sign in each direction.
The infinite board consists of all squares (m, n) with m, n nonnegative integers.
Use mathematical induction to show that a knight starting at $(0, 0)$ can visit every square in a finite sequence of legal moves.
Hint from Rosen: use induction on $s = m + n$. Every visited square must stay on the board.